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(54) PORTABLE PAYMENT TERMINAL

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CPC ***G07G 1/0018*** (2013.01); ***G06Q 20/352*** (2013.01); ***G06Q 20/353*** (2013.01)

(57) **ABSTRACT**

A portable payment terminal includes: a touch panel including a display on a front of a housing; a slot into which a contact IC card is to be inserted; a slit through which a magnetic card is to be swiped; a lens being arranged on the front; and a speaker unit. The slot is arranged on a top of the housing and is within a width of the display, the slit is arranged across the top of the housing and two housing sides adjacent to the top, the slot and the slit are arranged substantially parallel to the front, an insertion direction of the contact IC card and a swipe direction of the magnetic card are arranged substantially perpendicular to each other, and the lens and the speaker unit are arranged closer to a bottom of the housing with respect to the display and within the width of the display.

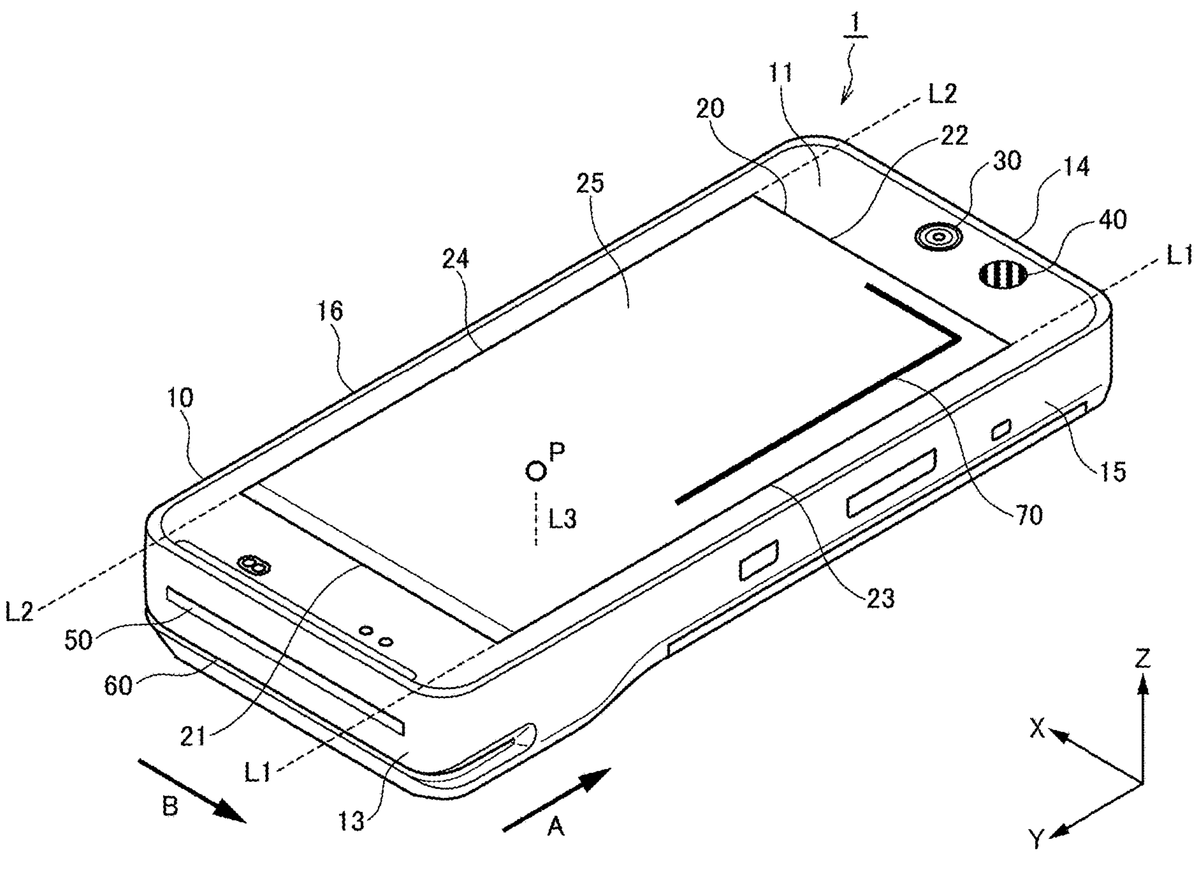


FIG. 1

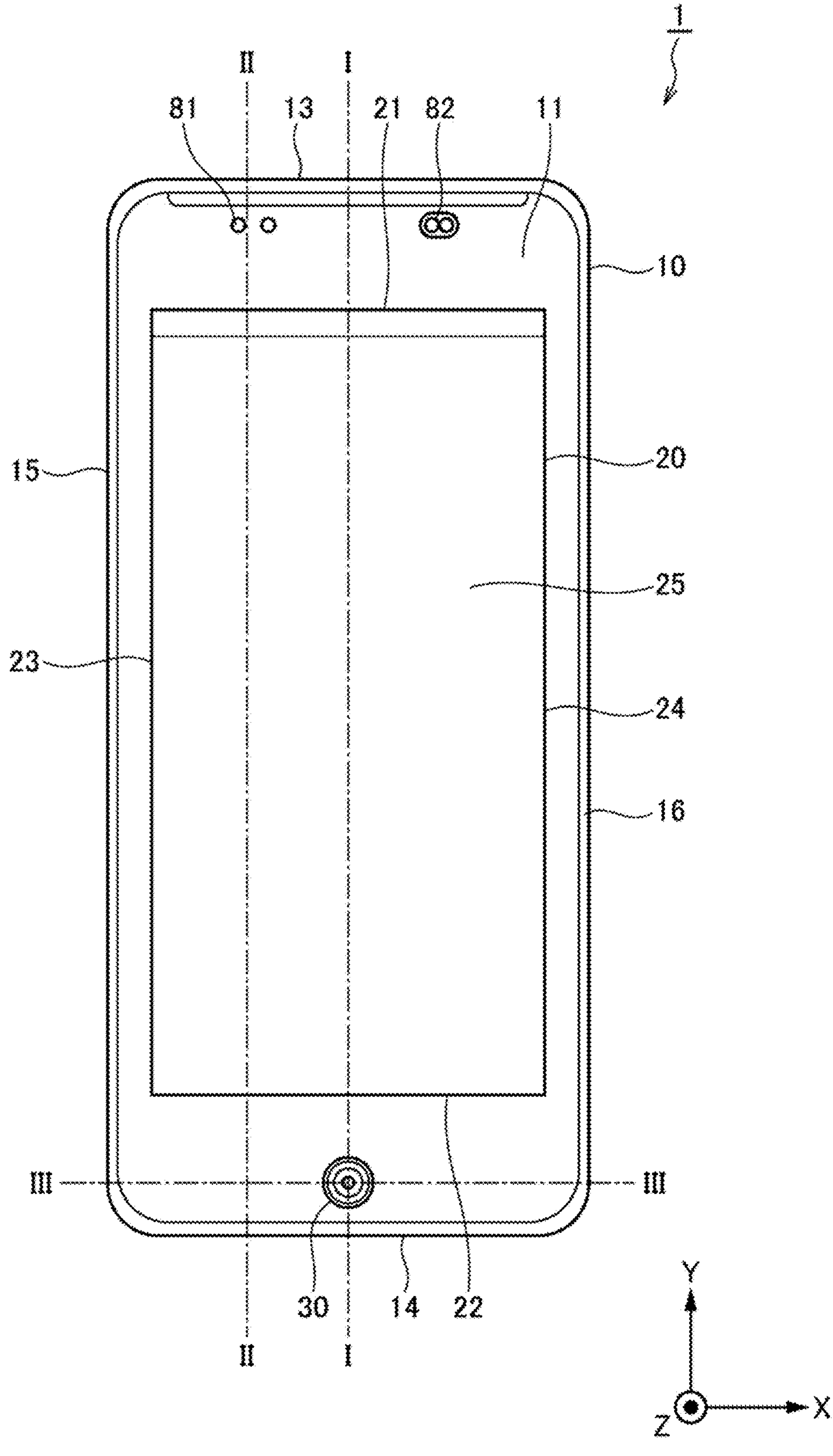


FIG. 2

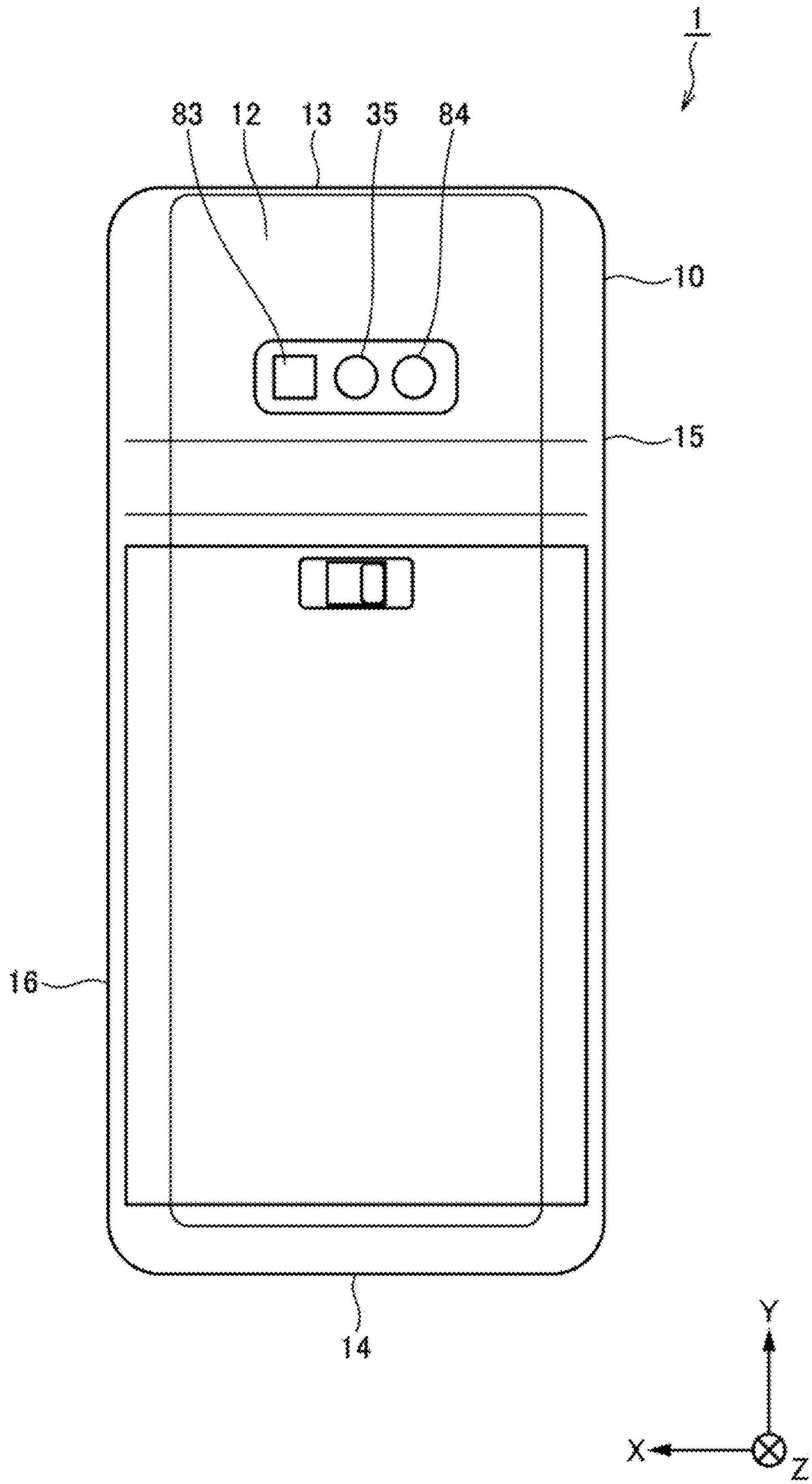


FIG. 3

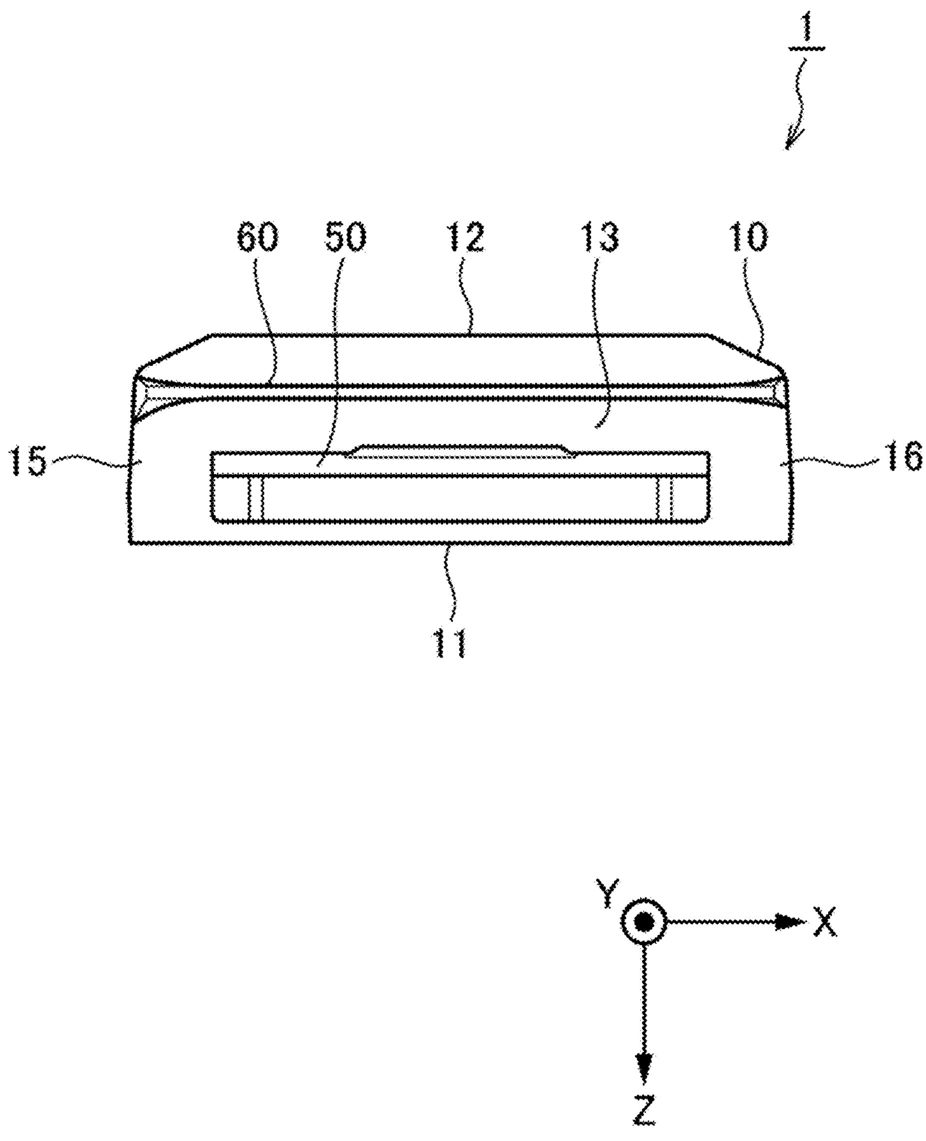


FIG. 4

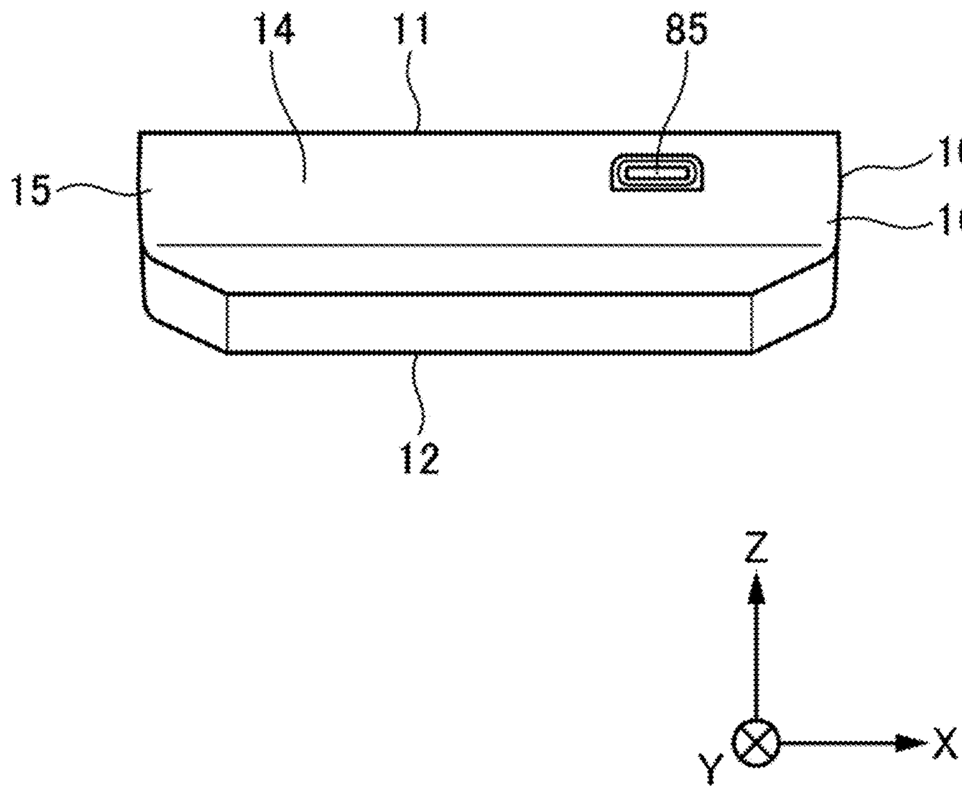


FIG. 5

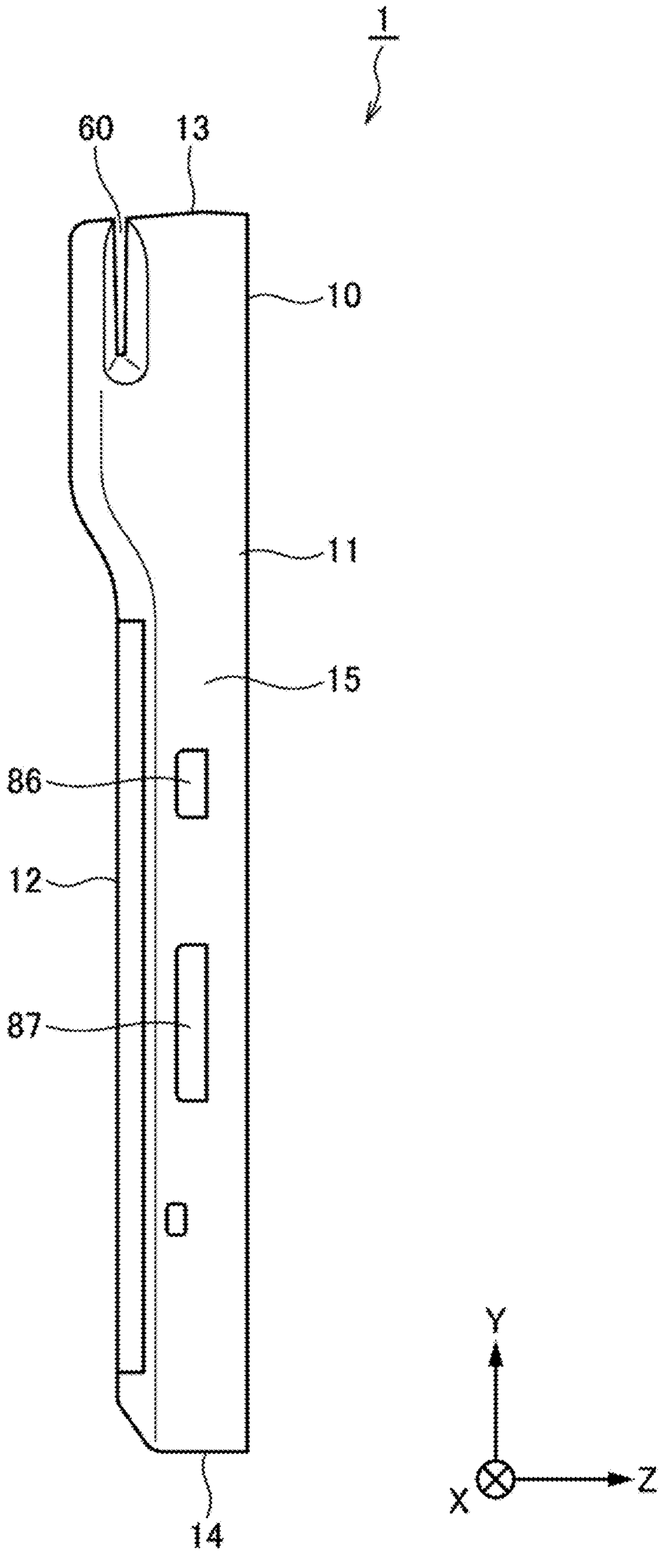


FIG. 6

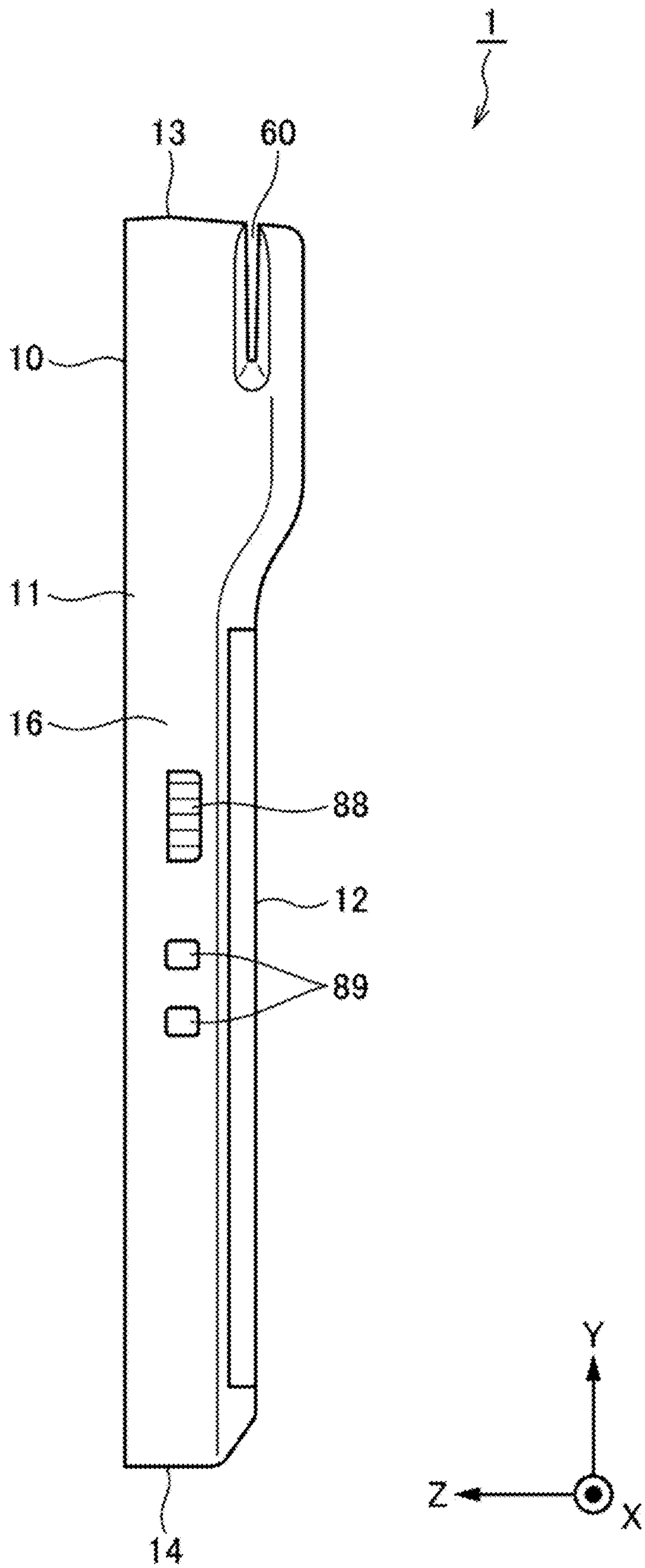


FIG. 7

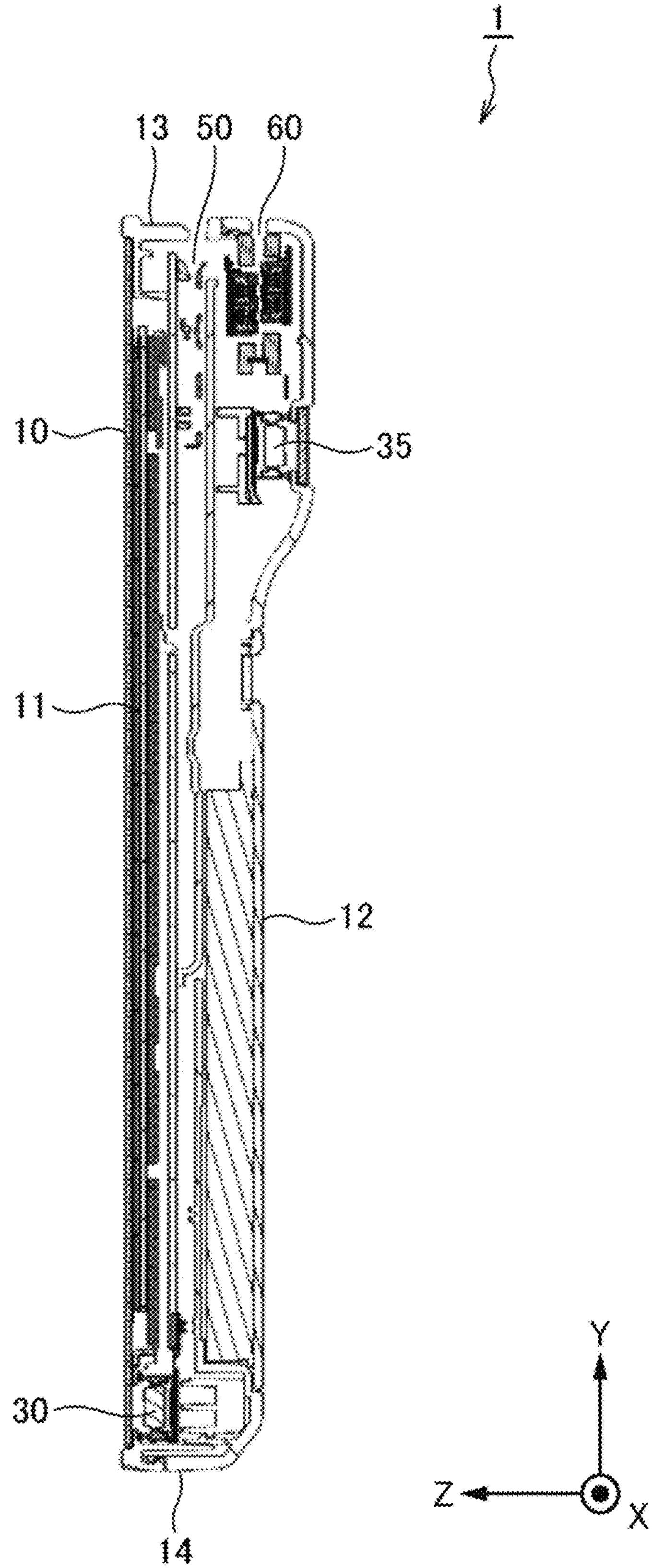


FIG. 8

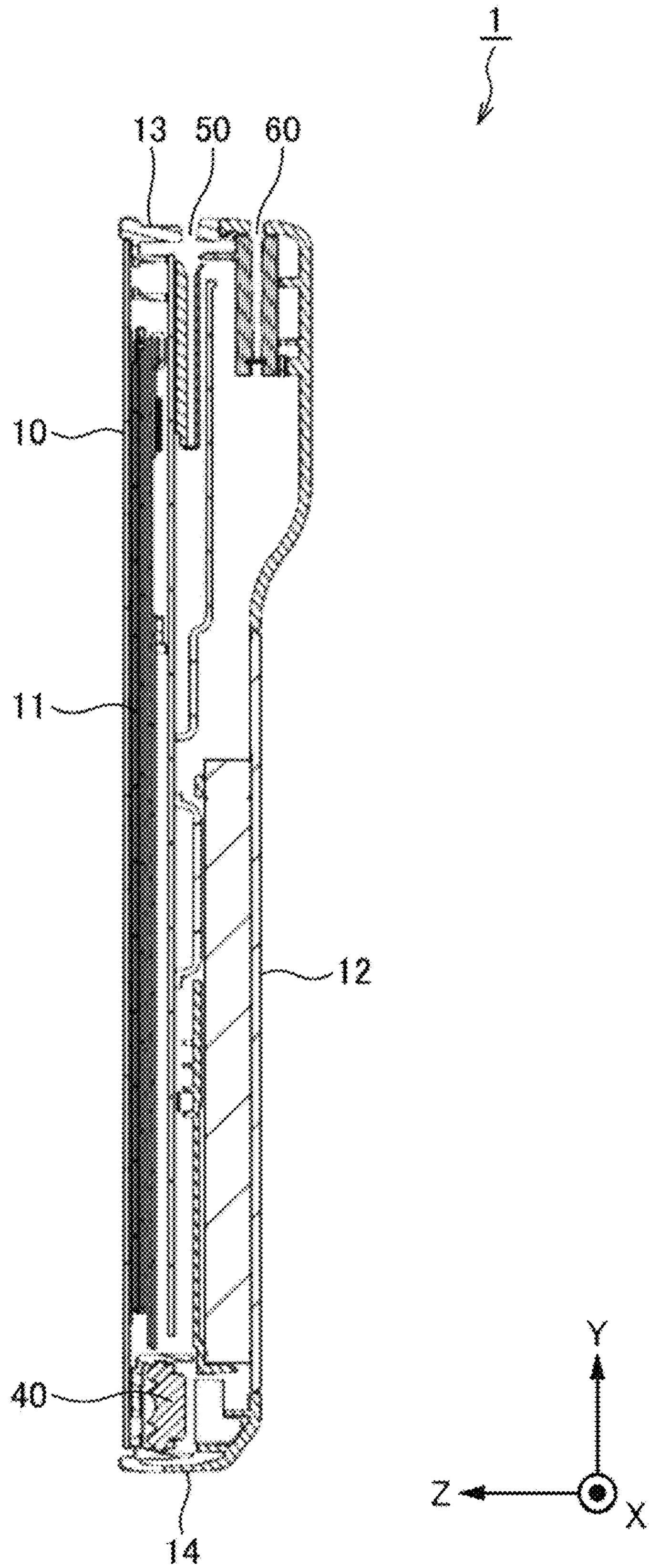


FIG. 9

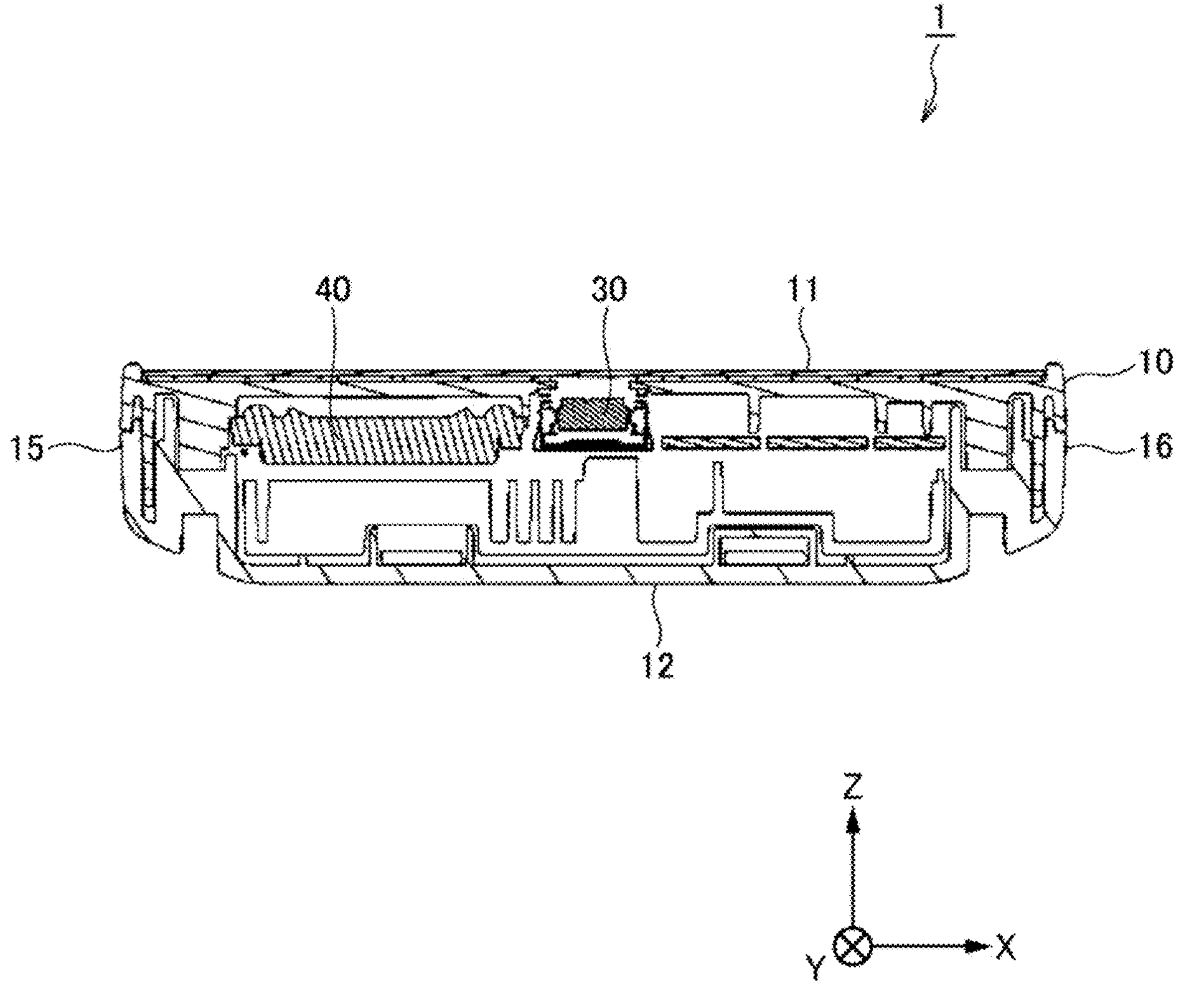
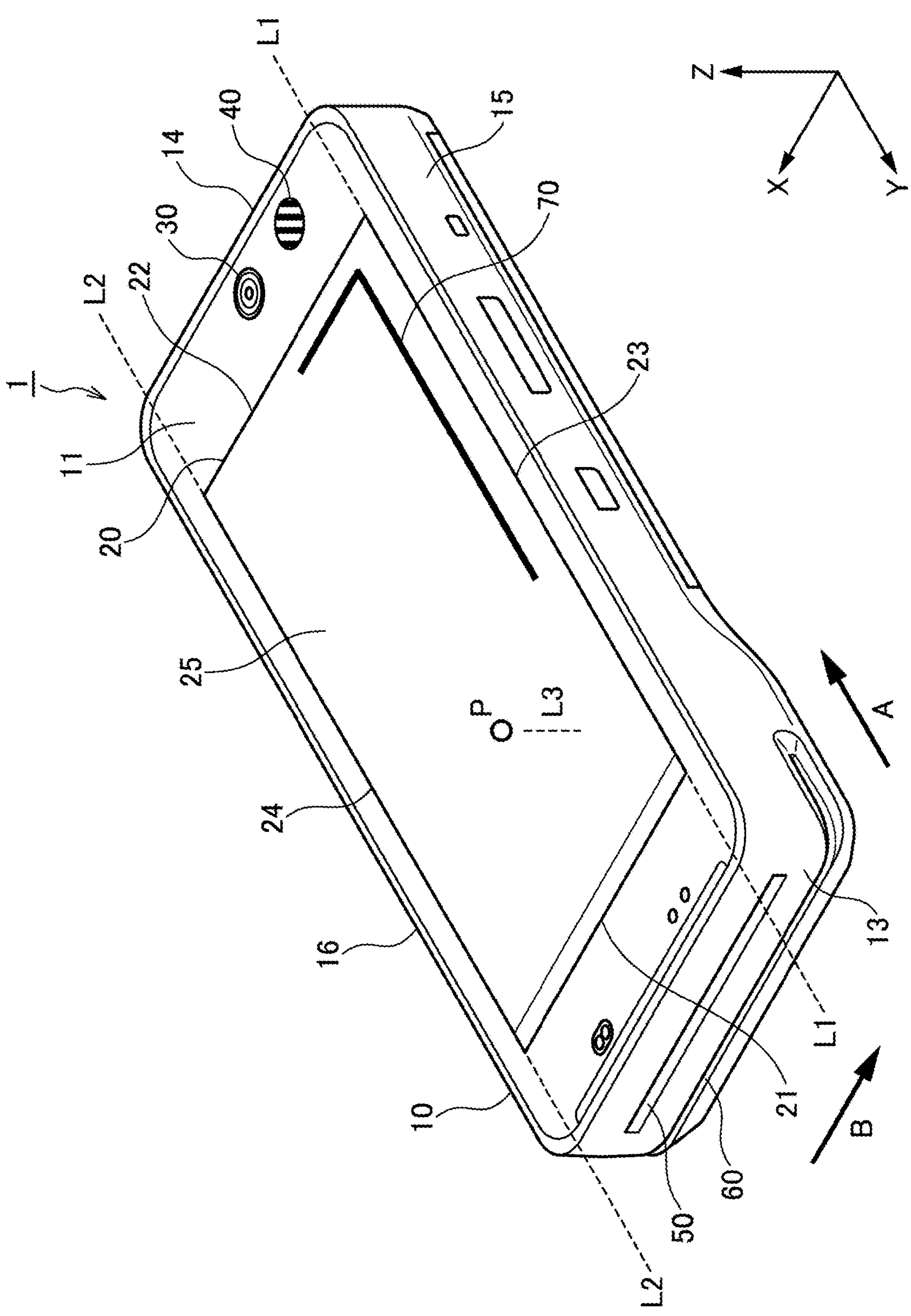


FIG. 10


1
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L2
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P
L3
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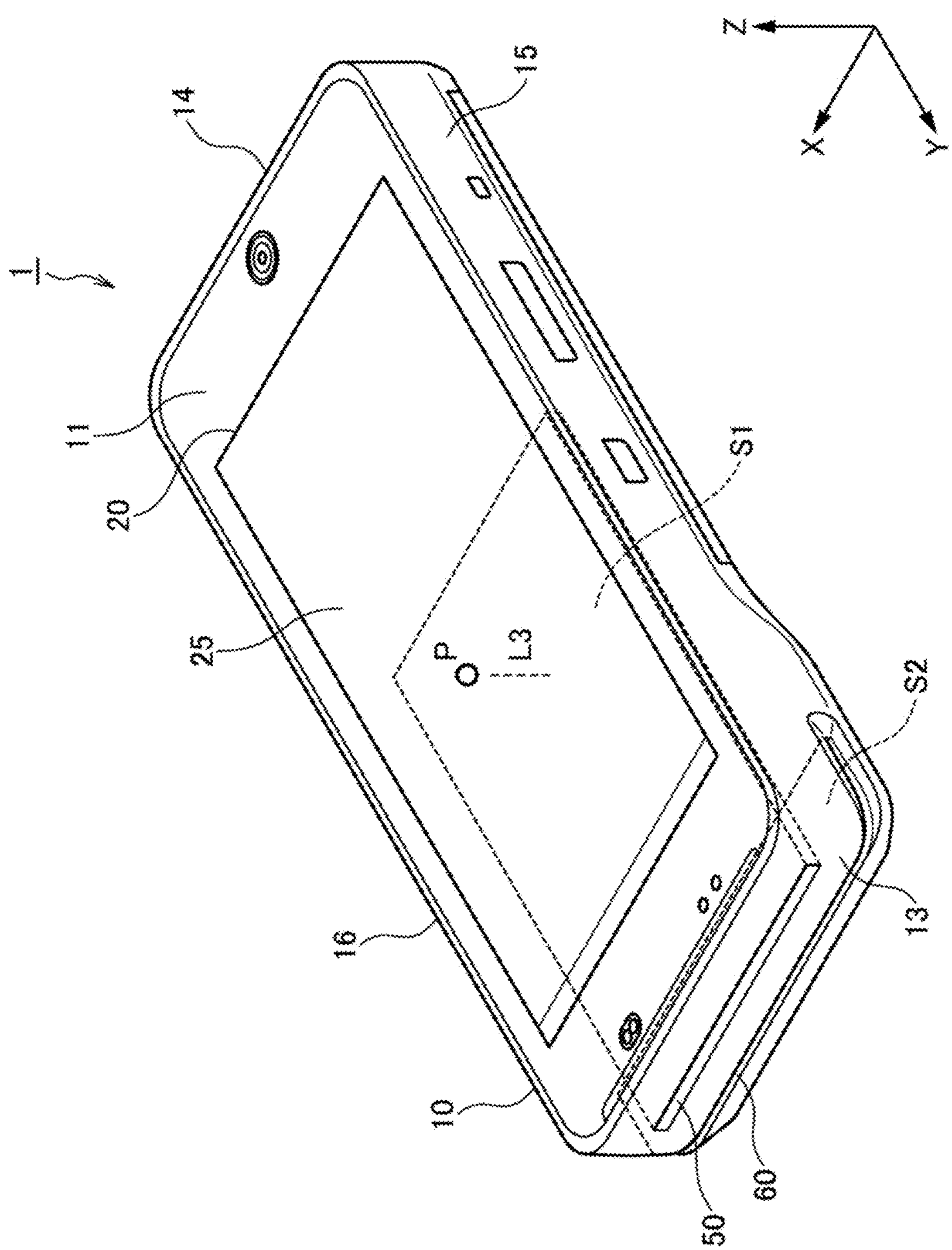
13
L1
50
60
L2
A
B
Z
X
Y

FIG. 11



PORTABLE PAYMENT TERMINAL

BACKGROUND

1. Technical Field

[0001] The present disclosure relates to a portable payment terminal.

2. Description of the Related Art

[0002] As the payment terminal, for example, a device that is installed at a counter of a shop and performs payment is known. The payment terminal includes a touch panel, and can perform payment on the basis of personal information input from the touch panel. In addition, the payment terminal is also compatible with a payment medium such as a card, and can perform payment by reading personal information or the like from such a medium.

[0003] With the diversification of transactions in recent years, the place of payment is not limited to a store counter, but has been expanded to various places. For example, this is a case where payment for a home delivery article is made at a delivery destination. For this reason, a portable payment terminal that can be carried is more emphasized (see Patent Literatures (PTLs) 1 to 3).

[0004] PTL 1: Unexamined Japanese Patent Publication No. 2019-144944 A

[0005] PTL 2: Unexamined Japanese Patent Publication No. 2012-185543 A

[0006] PTL 3: Unexamined Japanese Patent Publication No. 2017-228237 A

SUMMARY

[0007] The portable payment terminal is required to be downsized for carrying. However, there are various types of payment such as code payment, contactless payment, payment by a contact IC card, and payment by a magnetic card, and the device used differs depending on the type of payment. When the terminal is equipped with a plurality of different devices, downsizing is more difficult. In particular, a component (hereinafter, referred to as a tall component) having a tall portion (long in the thickness direction of the terminal) such as a camera, a speaker, or the like increases the thickness of the terminal, and can be an obstacle to miniaturization.

[0008] The present disclosure provides a portable payment terminal capable of making payment by various payment methods and achieving downsizing, particularly thinning.

[0009] A portable payment terminal of the present disclosure includes: a touch panel including a display, the touch panel being on a front of a housing; an antenna used for contactless payment; a slot into which a contact IC card for payment processing is to be inserted; a slit through which a magnetic card for the payment processing is to be swiped; a lens of a first camera that images a code of the payment processing, the lens being arranged on the front; and a speaker unit that is in the housing and outputs a sound of the payment processing. The slot is arranged on a top of the housing and is within a width of the display, the slit is arranged across the top of the housing and two housing sides adjacent to the top, the slot and the slit are arranged substantially parallel to the front of the housing, an insertion direction of the contact IC card into the slot and a swipe direction of the magnetic card into the slit are arranged substantially perpendicular to each other, and the lens and the speaker unit are arranged closer to a bottom of the housing with respect to the display and within the width of the display.

[0010] According to the present disclosure, payment can be made by various payment methods, and downsizing, particularly thinning, of a portable payment terminal can be achieved.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a front view illustrating a configuration example of a portable payment terminal according to an exemplary embodiment of the present disclosure;

[0012] FIG. **2** is a rear view illustrating a configuration example of the portable payment terminal;

[0013] FIG. **3** is a top view illustrating a configuration example of the portable payment terminal;

[0014] FIG. **4** is a bottom view illustrating a configuration example of the portable payment terminal;

[0015] FIG. **5** is a left side view illustrating a configuration example of the portable payment terminal;

[0016] FIG. **6** is a right side view illustrating a configuration example of the portable payment terminal;

[0017] FIG. **7** is a cross-sectional view taken along line I-I in FIG. **1**;

[0018] FIG. **8** is a cross-sectional view taken along line II-II in FIG. **1**;

[0019] FIG. **9** is a cross-sectional view taken along line III-III in FIG. **1**;

[0020] FIG. **10** is a first perspective view illustrating a schematic configuration example of the portable payment terminal; and

[0021] FIG. **11** is a second perspective view illustrating a schematic configuration example of the portable payment terminal.

DETAILED DESCRIPTIONS

[0022] Exemplary embodiments will now be described below in detail, referring to the drawings as appropriate. However, unnecessarily detailed description may be omitted. For example, a detailed description of a well-known matter and a repeated description of substantially the same configuration may be omitted. This is to avoid unnecessary redundancy of the following description and to facilitate understanding of those skilled in the art. Note that the accompanying drawings and the following description are provided for those skilled in the art to fully understand the present disclosure, and are not intended to limit the subject matter described in the claims.

[0023] FIG. **1** is a front view illustrating a configuration example of portable payment terminal **1** (mobile payment terminal) according to an exemplary embodiment of the present disclosure. FIG. **2** is a rear view illustrating a configuration example of portable payment terminal **1**. FIG. **3** is a top view illustrating a configuration example of portable payment terminal **1**. FIG. **4** is a bottom view illustrating a configuration example of portable payment terminal **1**. FIG. **5** is a left side view illustrating a configuration example of portable payment terminal **1**. FIG. **6** is a right side view illustrating a configuration example of portable payment terminal **1**.

[0024] Portable payment terminal **1** is a composite type in which a customer can designate a plurality of payment methods. Portable payment terminal **1** is a mobile terminal that can be held and used by a user including a customer and a store clerk who prompts the customer to make payment.

[0025] The plurality of payment methods include at least code payment, contactless payment, contact IC card payment, and magnetic card payment. The code payment is payment by reading a predetermined code, and includes barcode payment, two-dimensional code (for example, QR code (registered trademark)) payment, and the like. The contactless payment is payment made by using contactless communication (for example, near field communication (NFC)), and includes electronic money payment, contactless IC card payment, and the like. The contact IC card payment is payment performed by directly contacting the IC chip of the contact IC card with the contact of the reader/writer, and includes cash card payment, credit card payment, and the like. The magnetic card payment is payment for recording data in a magnetic body of a magnetic card, and includes credit card payment and the like. The plurality of payment methods are performed using each of a plurality of corresponding payment interfaces (described later). Note that portable payment terminal **1** may support other payment methods.

[0026] Portable payment terminal **1** has a housing **10** having, for example, a generally box shape. Housing **10** is formed of, for example, resin or the like, and is gripped by the user. Housing **10** has front **11** (FIG. **1**) mainly visually recognized by the user and rear **12** (FIG. **2**) facing front **11**. Further, housing **10** has top **13** (FIG. **3**) and bottom **14** (FIG. **4**) arranged between front **11** and rear **12**. During normal use of portable payment terminal **1**, the user places top **13** on the side far from the user himself/herself while looking at front **11**, faces bottom **14** toward the user side, and grips the vicinity of bottom **14** with his/her hand to use portable payment terminal **1**. Therefore, bottom **14** is a surface on the user side, and top **13** is a surface on the side opposite to the user. Further, housing **10** has two housing sides arranged between front **11** and rear **12**, specifically, left housing side **15** (FIG. **5**) and right housing side **16** (FIG. **6**). Top **13**, bottom **14**, left housing side **15**, and right housing side **16** are substantially perpendicular to front **11** and rear **12**.

[0027] Note that the coordinate axes in the drawing include, on a plane parallel to front **11**, an X-axis that is a direction parallel to top **13** and bottom **14** of the housing, and a Y-axis that is perpendicular to the X-axis and parallel to left housing side **15** and right housing side **16**. The coordinate axes further include a Z-axis that is perpendicular to the X-axis and the Y-axis. The X-axis direction corresponds to the width direction of portable payment terminal **1**, the Y-axis direction corresponds to the longitudinal direction of portable payment terminal **1**, and the Z-axis direction corresponds to the thickness direction of portable payment terminal **1**.

[0028] As illustrated in FIG. **1**, portable payment terminal **1** includes power supply/operation indicating lamp **81** exposed from front **11**, illuminance sensor **82** exposed from front **11**, touch panel **20** exposed from front **11**, and first camera **30** exposed from front **11**. As illustrated in FIG. **2**, portable payment terminal **1** includes second camera **35** exposed from rear **12**, light **83** exposed from rear **12**, and aimer **84** exposed from rear **12**. Details of touch panel **20**, first camera **30**, and second camera **35** will be described later.

[0029] As illustrated in FIG. **3**, portable payment terminal **1** includes contact IC card slot **50** which is arranged on top **13** and into which a contact IC card is inserted. As illustrated in FIGS. **3**, **5**, and **6**, portable payment terminal **1** includes magnetic card slit **60** which is arranged across top **13** and two housing sides (left housing side **15** and right housing side **16**) for swiping the magnetic card. As illustrated in FIG. **4**, portable payment terminal **1** further includes USB port **85**. Details of contact IC card slot **50** and magnetic card slit **60** will be described later.

[0030] As illustrated in FIGS. **5** and **6**, portable payment terminal **1** includes power supply button **86** and volume button **87** exposed from left housing side **15**, side button **88** exposed from right housing side **16**, and charging terminals **89**.

[0031] As described above, portable payment terminal **1** includes touch panel **20**. Touch panel **20** has display **25** that is exposed from front **11**, can be visually recognized by the user, and can be operated by finger contact, proximity, or the like. Display **25** has a quadrangular shape arranged by top side **21** corresponding to top **13**, bottom side **22** corresponding to bottom **14**, left side **23** corresponding to one housing side (left housing side **15**), and right side **24** corresponding to the other housing side (right housing side **16**).

[0032] FIG. **7** is a cross-sectional view taken along line I-I in FIG. **1**. First camera **30** is arranged closer to bottom **14** in housing **10**, the camera lens of first camera **30** is exposed from front **11**, and can image a code of code payment arranged to face front **11**. The portion having the camera lens is a tall portion of first camera **30**. In portable payment terminal **1**, a processor (not illustrated) can read the code and recognize the content of the code on the basis of the captured image. For example, when the user grips portable payment terminal **1** with the left hand and holds the code displayed on the screen of his/her own smartphone gripped with the right hand over first camera **30**, first camera **30** images the code, and the processor can read the code on the basis of the captured image and perform code payment.

[0033] Portable payment terminal **1** further includes second camera **35**. Second camera **35** is arranged closer to top **13** in housing **10**, and the camera lens of second camera **35** is exposed from rear **12**. Second camera **35** is a camera for imaging a commodity code on a commodity from rear **12**, separately from the payment described above. In portable payment terminal **1**, a processor (not illustrated) can read a commodity code on the basis of a captured image and recognize content related to the commodity code. For example, the user grips portable payment terminal **1** and turns rear **12** to the commodity side, second camera **35** images the code printed on the packing box of the commodity, and the processor can read the commodity code on the basis of the captured image to acquire information of the commodity.

[0034] FIG. **8** is a cross-sectional view taken along line II-II of FIG. **1**. FIG. **9** is a cross-sectional view taken along line III-III of FIG. **1**. Portable payment terminal **1** further includes speaker **40**. Speaker **40** is arranged closer to bottom **14** in housing **10** similarly to first camera **30**, and the speaker hole is exposed from front **11**. Speaker **40** can output a sound of the payment processing from front **11** at the time of payment. A speaker unit that is a portion from which sound is output is a tall portion of speaker **40**. The speaker hole is arranged at a position facing the speaker unit.

[0035] FIG. **10** is a first perspective view illustrating a schematic configuration example of portable payment terminal **1**. FIG. **11** is a second perspective view illustrating a schematic configuration example of portable payment terminal **1**. FIG. **10** is a diagram mainly illustrating an external configuration of portable payment terminal **1**. FIG. **11** is a diagram mainly illustrating an internal configuration of portable payment terminal **1**. Note that FIGS. **10** and **11** schematically illustrate portable payment terminal **1** according to the exemplary embodiment, and portable payment terminal **1** according to the exemplary embodiment is different in shape from portable payment terminal **1** illustrated in FIGS. **1** to **9**, but is the same terminal as portable payment terminal **1** illustrated in FIGS. **1** to **9**.

[0036] As illustrated in FIG. **10**, first extension line L**1** and second extension line L**2** which are two extension lines are virtually defined in order to specify portable payment terminal **1** according to the exemplary embodiment. First extension line L**1** is a line obtained by extending left side **23** of display **25** of touch panel **20** which is linear. Second extension line L**2** is a line obtained by extending right side **24** of display **25** of touch panel **20** which is linear.

[0037] The camera lens of first camera **30** is arranged in a region (that is, within the width of display **25**) between first extension line L**1** and second extension line L**2** in housing **10** and closer to bottom **14** of housing **10** with respect to bottom side **22** of display **25**. The speaker unit that outputs the sound of speaker **40** is also arranged in a region between first extension line L**1** and second extension line L**2** in housing **10** and closer to bottom **14** of housing **10** than bottom side **22** of display **25**. The speaker hole of the speaker unit is also arranged in a region between first extension line L**1** and second extension line L**2** and closer to bottom **14** of housing **10** than bottom side **22** of display **25**. Therefore, the appearance of the portable payment terminal including the camera lens of the first camera and the speaker hole can be made compact.

[0038] Contact IC card slot **50** is arranged in a region between first extension line L**1** and second extension line L**2** and on top **13** of housing **10**. Magnetic card slit **60** is arranged across top **13** and two housing sides (left housing side **15** and right housing side **16**) of housing **10**.

[0039] Furthermore, portable payment terminal **1** includes antenna **70** arranged along at least one of top side **21**, bottom side **22**, left side **23**, and right side **24** of display **25** in housing **10** and used for contactless payment. Antenna **70** of the present exemplary embodiment is an L-shaped antenna arranged along left side **23** and bottom side **22**, but may be an I-shaped antenna or an antenna having another shape.

[0040] As illustrated in FIG. **11**, first space S**1** arranged inside contact IC card slot **50** and second space S**2** arranged inside magnetic card slit **60** are arranged in housing **10**. First space S**1**, second space S**2**, and front **11** of housing **10** are arranged substantially parallel to each other.

[0041] In addition, first space S**1** and second space S**2** are arranged closer to top **13** of housing **10** than vertical line L**3** extending from center P of display **25** in front view toward rear **12** of housing **10**. Contact IC card slot **50** and first space S**1** are arranged such that insertion direction A of the contact IC card is substantially parallel to first extension line L**1** and second extension line L**2**. Magnetic card slit **60** and second space S**2** are arranged such that swipe direction B of the magnetic card is perpendicular to each of first extension line L**1** and second extension line L**2**.

[0042] Components such as first camera **30** and speaker **40** are relatively tall components as compared with other components such as power supply/operation indicator lamp **81**, illuminance sensor **82**, light **83**, aimer **84**, or USB port **85**, and when arranged in housing **10**, portable payment terminal **1** tends to increase in size, particularly in the thickness direction (Z-axis direction). However, in the present exemplary embodiment, components such as first camera **30** and speaker **40** that are relatively tall are arranged closer to bottom **14** of housing **10**, and contact IC card slot **50** and magnetic card slit **60** are arranged closer to top **13** of housing **10** where these tall components are not present. Therefore, it is possible to suppress an increase in size, particularly an increase in thickness, of portable payment terminal **1** and to easily achieve a reduction in thickness.

[0043] In addition, the camera lens of second camera **35** is arranged in a region between first extension line L**1** and second extension line L**2** in housing **10** and closer to top **13** of housing **10** than vertical line L**3**.

[0044] Also in second camera **35**, a portion having a camera lens is a relatively tall component. As illustrated in FIGS. **5** and **6**, the thickness on the side of top **13** of housing **10** on which second camera **35** is arranged is larger than the thickness on the side of bottom **14**, but the user does not grip the side of top **13** of housing **10** in normal use. As a result, it is possible to reduce the thickness of bottom **14** side of housing **10** gripped by the user and improve the handleability of portable payment terminal **1** while also mounting second camera **35** capable of reading the commodity code.

[0045] As described above, in portable payment terminal **1** of the present exemplary embodiment, components such as first camera **30** and speaker **40** having a relatively tall portion are arranged closer to bottom **14** of housing **10**. In addition, contact IC card slot **50** and magnetic card slit **60** are arranged closer to top **13** of housing **10** where a component having such a tall portion does not exist, that is, are arranged in the same direction while being concentrated in one direction. Therefore, the user can easily use contact IC card slot **50** and magnetic card slit **60** while gripping housing **10**. In addition, it is possible to easily reduce the thickness of portable payment terminal **1**.

[0046] In addition, since first camera **30** is arranged closer to bottom **14**, it is difficult for the user to block the display on display **25** by hand, and it is easy for the user to hold the code over first camera **30** in a state where the user can easily visually recognize display **25**. In addition, since portable payment terminal **1** includes both first camera **30** and second camera **35**, it is possible to perform both commodity management and payment.

[0047] As described above, at least the following matters are described in the present disclosure. Note that the components and the like corresponding to the above-described exemplary embodiments are shown in parentheses, but the present invention is not limited thereto.

[Item 1]

[0048] A portable payment terminal (portable payment terminal **1**), including:

- **[0049]** a touch panel (touch panel **20**) including a display (display **25**), the touch panel being on a front (front **11**) of a housing (housing **10**);
- **[0050]** an antenna (antenna **70**) used for contactless payment;

[0051] a slot (contact IC card slot **50**) into which a contact IC card for payment processing is to be inserted;

[0052] a slit (magnetic card slit **60**) through which a magnetic card for the payment processing is to be swiped;

[0053] a lens of a first camera (first camera **30**) that images a code of the payment processing, the lens being arranged on the front; and

[0054] a speaker unit (portion that outputs the sound of speaker **40**) that is in the housing and outputs a sound of the payment processing,

[0055] in which

[0056] the slot is arranged on a top (top **13**) of the housing and is within a width of the display,

[0057] the slit is arranged across the top of the housing and two housing sides (left housing side **15**, right housing side **16**) adjacent to the top,

[0058] the slot and the slit are arranged substantially parallel to the front of the housing,

[0059] the slot and the slit are arranged such that an insertion direction of the contact IC card into the slot and a swipe direction of the magnetic card into the slit are arranged substantially perpendicular to each other, and

[0060] the lens and the speaker unit are arranged closer to a bottom (bottom **14**) of the housing with respect to the display and within the width of the display.

[Item 2]

[0061] The portable payment terminal according to Item 1, in which

[0062] the slot includes a first space (first space S**1**) in the slot,

[0063] the slit includes a second space (second space S**2**) in the slit, and

[0064] the first space and the second space are arranged substantially parallel to the front.

[Item 3]

[0065] The portable payment terminal according to Item 1 or 2, in which

[0066] the first space and the second space are arranged closer to the top of the housing than a center of the display.

[Item 4]

[0067] The portable payment terminal according to any one of Items 1 to 3, in which

[0068] the antenna is arranged along at least one side (top side **21**, bottom side **22**, left side **23** or right side **24**) of the display.

[0069] According to each of these items, in the portable payment terminal, components such as the first camera and the speaker having a relatively tall portion are arranged closer to the bottom of the housing, and the slot for the contact IC card and the slit for the magnetic card are arranged closer to the top of the housing where these tall components are not present. In addition, it is possible to easily reduce the thickness of portable payment terminal.

[Item 5]

[0070] The portable payment terminal according to any one of Items 1 to 4, further including a second camera (second camera **35**) that images a commodity code related to a commodity,

[0071] in which a lens of the second camera is arranged on a rear (rear **12**) of the housing and is arranged within the width of the display.

[0072] As a result, the portable payment terminal can perform commodity management by also including the second camera, and can also realize thinning of the bottom side of the housing gripped by the user.

[0073] Although various exemplary embodiments have been described above with reference to the drawings, it goes without saying that the present disclosure is not limited to such examples. It is obvious that those skilled in the art can conceive various changes or modifications within the scope described in the claims, and it is understood that those changes or modifications naturally belong to the technical scope of the present disclosure. In addition, the components in the above exemplary embodiments may be arbitrarily combined without departing from the gist of the invention.

[0074] The present disclosure is useful for a portable payment terminal or the like that can make payment by various payment methods and can achieve miniaturization, particularly thickness reduction.

What is claimed is:

1. A portable payment terminal, comprising:

a touch panel including a display, the touch panel being on a front of a housing;

an antenna used for contactless payment;

a slot into which a contact IC card for payment processing is to be inserted;

a slit through which a magnetic card for the payment processing is to be swiped;

a lens of a first camera that images a code of the payment processing, the lens being arranged on the front; and

a speaker unit that is in the housing and outputs a sound of the payment processing,

wherein

the slot is arranged on a top of the housing and is within a width of the display,

the slit is arranged across the top of the housing and two housing sides adjacent to the top,

the slot and the slit are arranged substantially parallel to the front of the housing,

an insertion direction of the contact IC card into the slot and a swipe direction of the magnetic card into the slit are arranged substantially perpendicular to each other, and

the lens and the speaker unit are arranged closer to a bottom of the housing with respect to the display and within the width of the display.

2. The portable payment terminal according to claim **1**, wherein

the slot includes a first space in the slot,

the slit includes a second space in the slit, and

the first space and the second space are arranged substantially parallel to the front.

3. The portable payment terminal according to claim **2**, wherein

the first space and the second space are arranged closer to the top of the housing than a center of the display.

4. The portable payment terminal according to claim **3**, wherein

the antenna is arranged along at least one side of the display.

5. The portable payment terminal according to claim **4**, further comprising a second camera that images a commodity code related to a commodity,

wherein a lens of the second camera is arranged on a rear of the housing and is arranged within the width of the display.

6. The portable payment terminal according to claim **2**, wherein

the antenna is arranged along at least one side of the display.

7. The portable payment terminal according to claim **6**, further comprising a second camera that images a commodity code related to a commodity,

wherein a lens of the second camera is arranged on a rear of the housing and is arranged within the width of the display.

8. The portable payment terminal according to claim **1**, wherein

the slot includes a first space in the slot,

the slit includes a second space in the slit, and

the first space and the second space are arranged closer to the top of the housing than a center of the display.

9. The portable payment terminal according to claim **8**, wherein

the antenna is arranged along at least one side of the display.

10. The portable payment terminal according to claim **9**, further comprising a second camera that images a commodity code related to a commodity,

wherein a lens of the second camera is arranged on a rear of the housing and is arranged within the width of the display.

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